

Metabolic Syndrome and Episodic Memory Performance in Older Adults

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Baseline participant characteristics were summarized overall and stratified by metabolic syndrome (MetS) status. Categorical variables were described using frequencies and percentages, and continuous variables were summarized using means and standard deviations or medians and interquartile ranges, as appropriate based on distributional characteristics. Group differences by MetS status were assessed using Pearson's chi-square tests for categorical variables. For continuous variables, two-sample *t*-tests were used when normality assumptions were met, and Wilcoxon rank-sum tests were applied when distributions were skewed or deviated from normality.

The primary outcome was episodic memory performance measured by the Selective Reminding Test (SRT) total score. The primary exposure was MetS status, modeled as a binary variable (yes vs. no). Linear regression models were used to estimate the association between MetS status and SRT performance, with SRT total score specified as the dependent variable. Linear regression was selected to quantify mean differences in SRT scores and to allow adjustment for potential confounders.

Secondary analyses examined other neuropsychological outcomes, including C-F-L Letter Fluency, Category Fluency, and Color Trails Tests 1 and 2, using analogous modeling approaches. Additional analyses evaluated metabolic burden using a continuous MetS risk score ranging from 0 to 5, modeled as the independent variable in linear regression to assess dose-response relationships with SRT performance.

To investigate whether individual metabolic abnormalities contributed differentially to memory performance, elevated blood pressure, elevated fasting glucose, high waist circumference,

elevated triglycerides, and low HDL cholesterol were entered simultaneously into multivariable linear regression models with SRT total score as the outcome.

Adjusted models included covariates selected *a priori* based on prior literature and course guidance, including age, sex, education, depressive symptoms (PROMIS), and physical activity (IPAQ total). Previous studies have demonstrated that these factors are associated with depreciated memory function and may confound the relationship between metabolic health and cognition. For all regression analyses, estimated regression coefficients (β), standard errors, and p-values were reported. Model-level significance was assessed using F-tests. All hypothesis tests were two-sided, and statistical significance was defined as $p < 0.05$.

Results

Study Population Characteristics

A total of 266 participants were included in unadjusted analyses; adjusted models used slightly fewer observations due to missing covariates/outcomes. Baseline characteristics by MetS status are summarized in Table 1. Participants were predominantly female and in their mid-60s, with a median SRT total score of ~39. Distributions of neuropsychological outcomes are shown in Figure 1; scores were generally unimodal, with greater spread for the Color Trails measures. Participants with MetS had lower SRT total and Category Fluency scores, while C–F–L Letter Fluency and Color Trails Test performance did not differ significantly by MetS status. Missingness was minimal overall, with modest missingness for education, physical activity, and Color Trails outcomes.

Primary Aim: Differences in SRT Scores by MetS Status

In unadjusted analyses, participants with metabolic syndrome (MetS) had significantly lower Selective Reminding Test (SRT) total scores than those without MetS. A Wilcoxon two-sample

test showed a significant difference in SRT score distributions by MetS status ($Z = -2.95$, $p = 0.0032$), with lower mean rank scores among participants with MetS (Table 2). In multivariable linear regression models adjusted for age, sex, education, depressive symptoms (PROMIS), and physical activity (IPAQ total), MetS remained significantly associated with lower SRT total scores. The adjusted model was statistically significant overall ($F = 11.02$, $p < 0.0001$), and MetS was an independent predictor of SRT performance (Type III SS $F = 6.93$, $p = 0.0090$), explaining approximately 20% of the variance ($R^2 = 0.205$; Table 2).

Secondary Aim: Other Neuropsychological Outcomes by MetS Status

In unadjusted analyses, Category Fluency scores differed significantly by MetS status, with participants with MetS demonstrating lower performance than those without MetS (Wilcoxon $Z = -2.29$, $p = 0.022$; Table 2). No statistically significant unadjusted differences were observed for C–F–L Letter Fluency, Color Trails Test 1, or Color Trails Test 2 (all $p > 0.20$). After adjustment for age, sex, education, depressive symptoms, and physical activity, MetS status was not significantly associated with performance on any of these neuropsychological outcomes (all adjusted $p > 0.10$), although several adjusted models remained statistically significant overall (Table 2).

Continuous MetS Risk Score and SRT Performance

When metabolic syndrome (MetS) was modeled as a continuous exposure, higher MetS risk score was significantly associated with lower Selective Reminding Test (SRT) total scores. In unadjusted linear regression, each one-unit increase in MetS risk score was associated with a 1.45-point decrease in SRT score ($\beta = -1.45$, $SE = 0.38$, $p = 0.0002$; $R^2 = 0.053$). This association remained significant after adjustment for age, sex, education, depressive symptoms, and physical activity (Type III $F = 12.00$, $p = 0.0006$), with the adjusted model explaining

approximately 22% of the variance in SRT scores (Table 2). The inverse association between MetS risk score and SRT performance is illustrated in Figure 2.

Individual MetS Components and SRT Performance

Individual MetS Components and SRT Performance

To assess whether specific metabolic abnormalities were differentially associated with episodic memory, all five MetS components were entered simultaneously into linear regression models with Selective Reminding Test (SRT) total score as the outcome (Table 3). In unadjusted analyses, elevated fasting glucose, elevated triglycerides, and elevated blood pressure were associated with lower SRT scores (all $p < 0.01$; Table 2). After adjustment for age, sex, education, depressive symptoms, and physical activity, elevated fasting glucose ($\beta = -3.20$, $p = 0.0055$) and elevated triglycerides ($\beta = -3.49$, $p = 0.0084$) remained independently associated with poorer SRT performance, while blood pressure showed a marginal association ($p = 0.082$). Waist circumference and low HDL cholesterol were not significantly associated. The adjusted model was statistically significant ($F = 7.70$, $p < 0.0001$) and explained 23% of the variance in SRT scores.

Adjusted Models and Covariates

All regression models were adjusted for age, sex, education, depressive symptoms (PROMIS), and physical activity (IPAQ total) based on prior evidence linking these factors to cognitive performance. After adjustment, metabolic syndrome (MetS) remained significantly associated with Selective Reminding Test (SRT) scores. Participants with categorical MetS had lower SRT scores ($\beta = -2.41$, $p = 0.0090$), and higher MetS risk score was associated with poorer SRT performance in continuous models ($\beta = -1.45$ per unit increase, $p = 0.0006$; Table 3). In contrast, adjusted associations between MetS and other neuropsychological outcomes were not

statistically significant (Table 2). Education and sex were consistently associated with SRT performance, whereas age, depressive symptoms, and physical activity were not. Fully adjusted models explained approximately 20–23% of the variance in SRT scores.

Discussion

In this study, metabolic syndrome (MetS) was consistently associated with poorer episodic memory performance, as measured by the Selective Reminding Test (SRT). Participants with categorical MetS had significantly lower SRT scores than those without MetS, and this association remained robust after adjustment for age, sex, education, depressive symptoms, and physical activity. Modeling MetS as a continuous risk score further strengthened these findings, revealing a clear dose–response relationship in which increasing metabolic burden was associated with progressively worse memory performance. Together, these results suggest that cumulative metabolic risk, rather than MetS classification alone, is meaningfully related to memory function in older adults.

In contrast, associations between MetS and other neuropsychological outcomes were weaker and largely attenuated after adjustment. Although Category Fluency differed by MetS status in unadjusted analyses, this relationship did not persist in adjusted models, and no significant associations were observed for letter fluency or Color Trails Test performance. These findings suggest that MetS may preferentially affect episodic memory processes, rather than executive function or processing speed, at least in this sample.

Examination of individual MetS components provided additional insight into potential mechanisms. Elevated fasting glucose and elevated triglycerides were independently associated with lower SRT scores after adjustment, whereas waist circumference and low HDL cholesterol were not. These results are consistent with prior evidence implicating dysglycemia and lipid

dysregulation in hippocampal dysfunction and cognitive decline. Elevated blood pressure showed a marginal association, suggesting a possible but weaker contribution.

Several limitations should be considered. The cross-sectional design precludes causal inference, and residual confounding remains possible. Missing data for some covariates and outcomes reduced sample sizes in adjusted analyses, though missingness was modest. Finally, results may not generalize beyond this predominantly female, older adult cohort.

Overall, these findings support a specific and robust association between metabolic health and episodic memory, highlighting the importance of metabolic risk factor management for preserving cognitive function in aging populations.

Discussion

This study found consistent evidence that poorer metabolic health is associated with lower episodic memory performance. Participants with metabolic syndrome (MetS) had significantly lower Selective Reminding Test (SRT) scores than those without MetS, and this association remained after adjustment for demographic, behavioral, and psychological covariates. Modeling MetS as a continuous risk score further demonstrated a dose–response relationship, with higher metabolic burden associated with progressively worse memory performance.

In contrast, associations between MetS and other neuropsychological outcomes were limited. Although category fluency differed by MetS status in unadjusted analyses, this association did not persist after adjustment, and no significant relationships were observed for letter fluency or processing speed measures. These findings suggest that MetS may be more strongly related to episodic memory than to other cognitive domains.

Among individual MetS components, elevated fasting glucose and triglycerides were independently associated with poorer SRT performance, whereas waist circumference and low

HDL cholesterol were not. This pattern aligns with prior evidence linking dysglycemia and lipid abnormalities to memory-related neural pathways.

Key limitations include the cross-sectional design, potential residual confounding, and limited generalizability due to an older, predominantly female sample. Future longitudinal studies are needed to clarify causal relationships and underlying biological mechanisms.

Tables and Figures

Table 1. Comparison of baseline characteristics between Mets and non-Mets.

Variable	Overall N=266	No MetS N=145	MetS N=121	P-value
Age	64 (61, 67)	64 (61, 66)	65 (62, 68)	0.0353
Depression	51 (45, 56)	51 (45, 55)	51 (38, 57)	0.0104
Education	11 (8, 12)	12 (8, 12)	10 (7, 12)	0.9376
Physical Activity	3279 (1155, 4878)	3720 (1584, 5118)	2772 (869, 4158)	0.9376
SRT Total Words	39 (33, 45)	39 (35, 46)	37 (31, 43)	0.0032
Letter Fluency	29 (15, 51)	29 (15, 56)	29 (15, 47)	0.2350
Category Fluency	6 (2, 18)	9 (2, 18)	6 (1, 13)	0.0222
Color Trails Test 1	21 (2, 50)	24 (4, 54)	21 (1, 42)	0.2364
Color Trails Test 2	8 (1, 42)	8 (1, 42)	7 (1, 38)	0.7821
Sex				
Male	75 (28%)	42 (29%)	33 (27%)	0.7600
Female	191 (72%)	103 (71%)	88 (73%)	0.7600
Blood Pressure				
High	117 (44%)	28 (19%)	89 (74%)	<.0001
Low	149 (56%)	117 (81%)	32 (26%)	<.0001
Glucose				
High	109 (41%)	23 (16%)	86 (71%)	<.0001
Low	157 (59%)	122 (84%)	35 (29%)	<.0001
Waist circumference				
High	75 (28%)	42 (29%)	33 (27%)	<.0001
Low	98 (37%)	84 (58%)	14 (12%)	<.0001
Triglycerides				
High	58 (22%)	13 (9%)	45 (37%)	<.0001
Low	208 (78%)	132 (91%)	76 (63%)	<.0001
HDL cholesterol				
High	153 (58%)	44 (30%)	109 (90%)	<.0001
Low	113 (43%)	101 (70%)	12 (10%)	<.0001

1. Median (IQR); n (%)

2. Wilcoxon rank sum test; Pearson's Chi-squared test

Table 2. Associations Between Metabolic Syndrome (MetS) Measures and Neuropsychological Test Performance

Outcome	Unadjusted p-value	Adjusted p-value	Model F-test p-value
SRT: Categorical MetS	0.0032	0.0090	<0.0001
SRT: MetS Risk Score	0.0002	0.0006	<0.0001
C–F–L Letter Fluency	0.2350	0.4078	0.1024
Category Fluency	0.0222	0.1401	0.0002
Color Trails Test 1	0.2364	0.4723	0.0450
Color Trails Test 2	0.7821	0.9545	0.2716
SRT: Blood Pressure	<0.01	0.082	<0.0001
SRT: Fasting Glucose	<0.01	0.0055	<0.0001
SRT: Waist	<0.01	0.5359	<0.0001
SRT: Triglycerides	<0.01	0.0084	<0.0001
SRT: HDL Cholesterol	<0.01	0.4999	<0.0001

Table 3. Multivariable Linear Regression Models for Selective Reminding Test (SRT) Performance

SRT Categorical MetS model			
Predictor	β (Estimate)	SE	p-value
MetS (Yes vs No)	−2.41	0.91	0.0090
Age	−0.23	0.49	0.6318
Female sex	3.36	3.71	0.3657
Education	1.15	0.44	0.0090
PROMIS (depression)	0.04	0.21	0.8766
Physical activity (IPAQ)	0.01	0.04	0.6554
SRT Continuous MetS risk score model			
Predictor	β (Estimate)	SE	p-value
MetS risk score (0–5)	−1.45	0.38	0.0006
Age	−0.41	0.25	0.1963
Female sex	5.33	1.13	<0.0001
Education	1.71	0.33	<0.0001
PROMIS (depression)	0.07	0.24	0.7685
Physical activity (IPAQ)	0.02	0.07	0.8048
Individual MetS components model			
Predictor	β (Estimate)	SE	p-value
Elevated blood pressure	−2.00	1.15	0.082

Elevated fasting glucose	-3.20	1.14	0.0055
High waist circumference	1.95	1.14	0.0868
Elevated triglycerides	-3.49	1.31	0.0084
Low HDL cholesterol	-0.94	1.15	0.4166

Figure 1. Distribution of Neuropsychological Test Scores

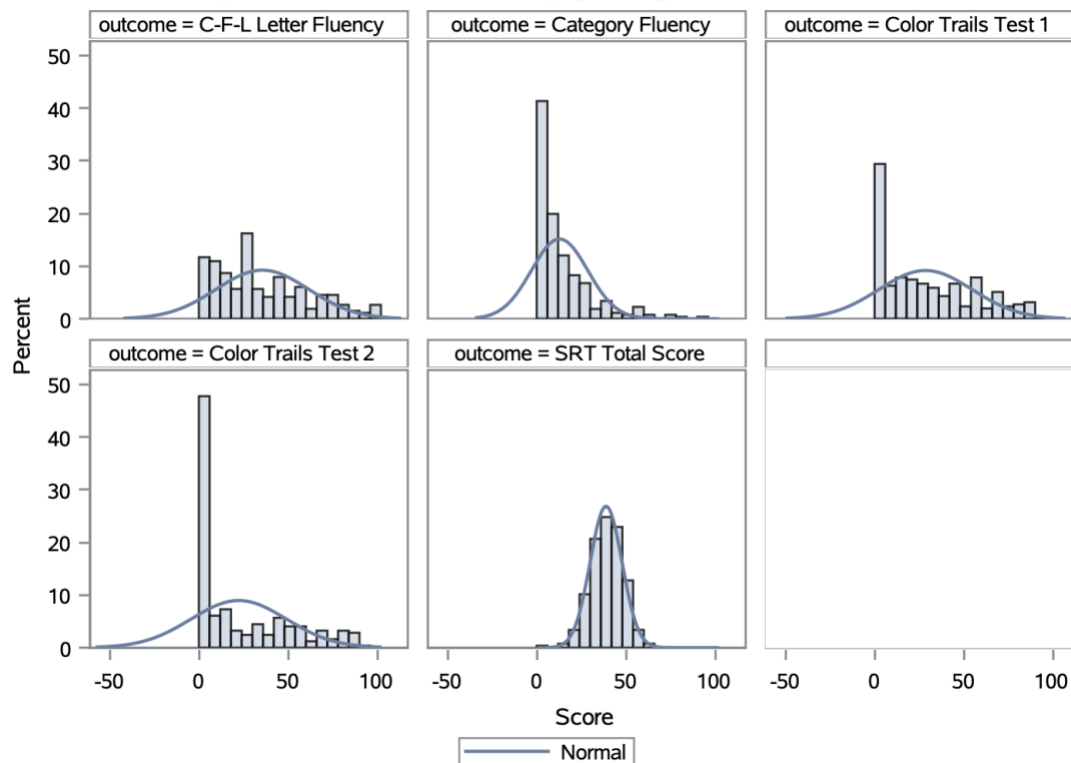


Figure 2. Selective Reminding Test Scores Across Metabolic Syndrome Risk Score Levels

